

Preservation bingo — Volunteer call sheet

Call a term and read the clue that fits the player age. Check it off as you go.

FOOD SAFETY

☐ **pressure canner** *(equipment)*

Kids: A big pot with a lid that locks tight and helps keep food safe by cooking it very hot

Tweens: A special pot that cooks food with steam at high pressure to kill bad bacteria like botulism

Teens: A sealed pot designed to reach temperatures above boiling to safely process low-acid foods and prevent botulism

Adult: A kettle with a clamped lid used to heat foods above boiling temperature (240°F) under steam pressure to destroy *Clostridium botulinum* and other pathogens in low-acid foods

Source: PNW 361 Canning meat poultry game

☐ **headspace** *(process)*

Kids: The empty space at the top of a jar before it is closed

Tweens: The space left between the food and the lid so food can cook safely inside the jar

Teens: The measured gap between food or liquid and the jar lid, crucial for proper heat penetration and sealing

Adult: The specific space (usually 1 inch) left between the top of food or liquid in a jar and the rim, ensuring vacuum sealing and proper heat transfer during canning

Source: SP 50-1000 Turkey basics

☐ **botulism** *(safety)*

Kids: A dangerous sickness that can happen if food is not cooked safely

Tweens: A harmful illness caused by bacteria that grow in canned foods that are not heated well

Teens: A severe foodborne illness caused by *Clostridium botulinum* toxins in improperly processed low-acid canned foods

Adult: A potentially fatal paralytic illness caused by ingestion of neurotoxins produced by *Clostridium botulinum* in anaerobic, low-acid, improperly processed home-canned foods

Source: PNW 361 Canning meat poultry game

☐ **pH** *(safety)*

Kids: A way to tell if something is sour or not

Tweens: A number that tells us how acidic or sour a food is, which helps keep it safe

Teens: A scale from 0 to 14 measuring acidity; foods with pH below 4.6 are high-acid and safer to water-bath can

Adult: A numeric measure of acidity or alkalinity; foods with pH $\leq$ 4.6 are high-acid and safe for water bath canning, while higher pH foods require pressure canning to prevent botulism

Source: SP 50-1000 Turkey basics

☐ **venting** *(process)*

Kids: Letting steam out before cooking to make food safe

Tweens: Releasing air from the canner by letting steam escape to get it ready for safe cooking

Teens: The process of releasing trapped air by letting steam flow continuously for 10 minutes before pressurizing a canner

Adult: Allowing steam to escape steadily from the petcock for 10 minutes before closing to remove non-condensable air, ensuring proper temperature and safety during pressure canning

Source: PNW 172 Canning vegetables

☐ **freezing** *(process)*

Kids: Putting food in the cold to keep it fresh and safe

Tweens: Making food very cold so tiny bugs can't grow and spoil it

Teens: Storing food at temperatures below 32°F to slow bacterial growth and preserve freshness safely

Adult: Preserving food by lowering temperature below 0°C (32°F), inhibiting microbial growth and enzymatic activity to maintain food safety and quality

Source: SP 50-1000 Turkey basics

☐ **pickling salt** *(ingredient)*

Kids: A special kind of salt used to make sour pickles

Tweens: Salt used in making pickles, without anything that can make cloudiness in the jar

Teens: Pure granulated salt without additives used in pickling to avoid jar cloudiness and help preservation

Adult: Pure sodium chloride salt without iodine or anti-caking agents to ensure clear brine and effective preservation in pickling processes

Source: SP 50-1000 Turkey basics

☐ **blanching** *(process)*

Kids: Quickly putting vegetables in hot water then cold water to keep them fresh

Tweens: A short heat step to stop enzymes that spoil vegetables, done before freezing or drying

Teens: Immersing vegetables briefly in boiling water or steam to stop enzymatic activity and preserve color, flavor, and nutrients before freezing or canning

Adult: A brief thermal treatment (boiling water or steam) to inactivate enzymes that cause spoilage, maintaining quality and safety during frozen or dried storage

Source: SP 50-1000 Turkey basics

☐ **seal check** *(safety)*

Kids: Making sure the jar lid is closed tight so no bugs get in

Tweens: Checking the jar lid after cooling to see if it is sealed properly and safe to store

Teens: Verifying jar lids curve downward and don't move when pressed, indicating a proper vacuum seal to prevent spoilage

Adult: Assessing after processing that jar lids have formed a vacuum seal, indicated by a downward concave lid center and no lid movement on pressure, ensuring safe preservation

Source: SP 50-1000 Turkey basics

CANNING FRUITS

☐ **jar lifter** *(equipment)*

Kids: A tool to safely grab hot jars.

Tweens: It helps you pick up hot jars without burning your hands.

Teens: A tool designed to grip and lift hot canning jars safely from boiling water.

Adult: A specialized gripping tool used to safely remove hot jars from boiling water canners or steam canners without risk of burns or dropping.

Source: PNW 172 Canning vegetables

☐ **ascorbic acid** *(ingredient)*

Kids: It's a special powder that helps fruit stay bright.

Tweens: A powder added to fruit to stop it from turning brown.

Teens: Vitamin C often used in canning to prevent browning and preserve color and freshness.

Adult: Ascorbic acid, also known as vitamin C, is used in canning fruits to prevent enzymatic browning and maintain color and nutritional quality.

Source: PNW 395 Salsa recipes for canning

☐ **hot pack** *(process)*

Kids: Putting hot fruit into jars.

Tweens: Cooking fruit lightly before putting it in jars.

Teens: A packing method where fruit is heated before filling jars, which helps preserve texture and color.

Adult: A canning method where fruit is heated in syrup or water before packing into jars, which reduces floating fruit and provides better color retention.

Source: PNW 199 Canning fruits

☐ **raw pack** *(process)*

Kids: Putting raw fruit in a jar.

Tweens: Packing uncooked fruit into jars with hot juice or water.

Teens: A method of filling jars with raw, uncooked fruit and covering with hot syrup or juice before processing.

Adult: A canning method where raw fruit is packed into jars and then covered with hot syrup, juice, or water for processing, resulting in firmer fruit texture.

Source: PNW 199 Canning fruits

☐ **pectin** *(ingredient)*

Kids: A jelly helper in fruit.

Tweens: A natural fiber in fruit that helps jelly set.

Teens: A plant-based carbohydrate that helps fruit spreads and jams thicken and gel.

Adult: A soluble fiber found in fruit cell walls that, when heated with sugar and acid, forms a gel network essential for proper jelly and jam consistency.

Source: PNW 172 Canning vegetables

☐ **boiling water canner** *(equipment)*

Kids: A big pot where jars get hot in water to keep food safe.

Tweens: A specially made pot to boil jars to keep canned food safe.

Teens: A kitchen vessel designed to hold jars submerged in boiling water for processing acidic foods safely.

Adult: A large pot with a lid and rack used to process high-acid foods in jars by boiling water at 212°F to destroy spoilage organisms and create a vacuum seal.

Source: PNW 199 Canning fruits

☐ **canning salt** (ingredient)

Kids: A special kind of salt used for canning.

Tweens: Salt used in canning that is pure and doesn't make cloudy jars.

Teens: Pure salt without additives, used to enhance flavor and preserve canned foods without causing cloudiness.

Adult: Fine-grained, additive-free salt used in home canning to season foods without producing haze or interfering with proper preservation.

Source: PNW 395 Salsa recipes for canning

☐ **air bubbles** (process)

Kids: Tiny air spots in jars that need to be made to go away.

Tweens: Air trapped inside jars that should be removed to make a good seal.

Teens: Air pockets trapped around food in jars that must be released to ensure even heat and proper seal.

Adult: Visual pockets of trapped air in jars that are removed using a plastic utensil to ensure accurate headspace measurement and prevent seal failure.

Source: PNW 199 Canning fruits

☐ **syrup** (ingredient)

Kids: A sweet, thick juice that covers fruit in jars.

Tweens: A sweet liquid used to keep canned fruit tasty and juicy.

Teens: A sugar and water mixture used in canning to preserve fruit flavor and texture.

Adult: A sugar solution of varying concentrations used as a packing liquid to preserve flavor, texture, and color of canned fruits.

Source: PNW 199 Canning fruits

CANNING TOMATOES & SALSAS

☐ **acidification** (safety)

Kids: Making tomatoes safe by adding sour things like lemon juice

Tweens: Adding vinegar or lemon juice to keep tomatoes safe from germs

Teens: Increasing acidity of tomatoes and salsas with lemon juice or vinegar to prevent harmful bacteria

Adult: The process of adding acid, such as lemon juice or citric acid, to tomatoes and tomato products to lower pH below 4.6 ensuring safe boiling water canning and preventing Clostridium botulinum growth

Source: PNW 300 Canning tomatoes and tomato products

☐ **altitude adjustment** (process)

Kids: Changing how long you cook because you live on a tall mountain

Tweens: Changing the cooking time or pressure when you live in high places

Teens: Increasing processing time or pressure in canning to compensate for lower boiling points at higher altitudes

Adult: Modifying pressure (e.g., weighted-gauge increase from 10 to 15 lbs) or processing time when canning above specific altitudes to ensure safe temperature and kill pathogens

Source: PNW 300 Canning tomatoes and tomato products

☐ **processing time** (process)

Kids: How long you need to cook jars to keep food safe

Tweens: The total minutes you have to cook canned foods to stop germs

Teens: The specific duration jars must be processed in the canner to ensure destruction of harmful microorganisms

Adult: The length of time jars must be held at a specified temperature and pressure in boiling water or pressure canning to achieve safe sterilization as per tested recipes

Source: PNW 300 Canning tomatoes and tomato products

☐ **steam canner** (equipment)

Kids: A big pot that uses steam to cook jars

Tweens: A canner that cooks jars with only steam instead of full water

Teens: A specialized canner using pure steam, needing less water than boiling water canners for high-acid foods under 45 minutes processing

Adult: A boiling water canning alternative that processes high-acid foods using pure steam; must use tested recipes and processing times under 45 minutes with altitude adjustments

Source: PNW 172 Canning vegetables

☐ **citric acid** *(ingredient)*

Kids: A sour powder like lemon that makes tomatoes safe

Tweens: A powder added to tomatoes to make them more acidic and safe to can

Teens: An acidulant added to tomato products to lower pH and prevent bacteria growth during canning

Adult: A commonly recommended food-grade acidifier added in measured amounts to tomatoes to ensure pH below 4.6 for safe boiling water canning

Source: PNW 300 Canning tomatoes and tomato products

☐ **jar cooling** *(process)*

Kids: Let jars rest quietly until they are not hot

Tweens: Leaving jars alone to cool after canning so they seal well

Teens: Allowing jars to cool undisturbed for 12 to 24 hours to ensure vacuum seal forms and prevent spoilage

Adult: Cooling jars on a rack or towel with air circulation, avoiding cold drafts, for 12 to 24 hours post-processing to complete sealing and preserve quality

Source: PNW 300 Canning tomatoes and tomato products

☐ **reprocessing** *(process)*

Kids: Cooking a jar again if it did not close right

Tweens: If a jar did not seal, cook it again soon to keep the food safe

Teens: The act of re-canning food within 24 hours using new lids if the jar did not seal properly initially

Adult: After a jar fails seal testing, reprocessing with appropriate packing method, new lid, and full processing time within 24 hours to ensure safety

Source: PNW 199 Canning fruits

☐ **lemon juice** *(ingredient)*

Kids: Sour juice from lemons that makes things safe

Tweens: Acid added to tomatoes to keep germs away

Teens: A natural acidifier commonly used in canning tomatoes to lower pH and ensure safe preservation

Adult: Bottled lemon juice standardized to 5% acidity, used in recommended amounts to acidify tomatoes and tomato products for safe boiling water canning

Source: PNW 300 Canning tomatoes and tomato products

☐ **siphoning** *(result)*

Kids: When the juice leaks out of the jar after cooking

Tweens: Loss of liquid from jars during cooling that can ruin the food

Teens: The undesirable loss of liquid from jars during or after processing, often caused by improper headspace or rapid cooling

Adult: The movement of liquid out of the jar during processing or cooling, leading to reduced liquid covering the food, possible spoilage, and compromised seal

Source: PNW 736 Fruit pie filling home canning

LOW ACID FOODS

☐ **exhausting** *(process)*

Kids: Letting out air from the pot before cooking to keep food safe.

Tweens: Removing air from a pressure canner by venting before pressure builds to make sure food cooks evenly.

Teens: The process of venting a pressure canner for about 10 minutes to remove trapped air for consistent high-temperature steam heat during canning.

Adult: The step of exhausting air by venting the pressure canner for 10 minutes before pressurizing to ensure steam replaces air for safe and consistent heating in low-acid food canning.

Source: PNW 361 Canning meat poultry game

☐ **rack** *(equipment)*

Kids: A metal thing you put jars on so they don't touch each other in the big pot.

Tweens: A flat circle or grid inside a canner that holds jars during processing so they don't sit directly on the bottom.

Teens: A removable metal rack placed in the canner to keep jars off the bottom and allow steam to circulate during pressure canning.

Adult: A flat wire or metal rack placed at the bottom and between layers of jars in a pressure canner to prevent direct contact with the base and improve even heat distribution.

Source: PNW 421 Use care operation of your pressure canner

☐ **pressure gauge** *(equipment)*

Kids: A part on the pot that tells how hard the steam is pushing inside.

Tweens: The dial or weight on a canner that shows or controls the pressure inside to keep food safe.

Teens: A device on a pressure canner indicating internal pressure (dial gauge) or controlling it by weight (weighted gauge) critical to reach safe temperatures for low-acid foods.

Adult: An instrument on a pressure canner, either dial-type for reading pressure or weighted gauge for regulating pressure, essential for maintaining the temperature above 240°F required to destroy *Clostridium botulinum* spores.

Source: PNW 421 Use care operation of your pressure canner

☐ **overfilling** (safety)

Kids: Putting too much food or liquid in the jar so it could be unsafe.

Tweens: Filling jars too full, which can cause them not to heat properly and be unsafe.

Teens: Packing jars beyond recommended headspace or contents interfering with heat penetration can cause under-processing and unsafe foods.

Adult: Over-packing jars reduces heat penetration, leading to inadequate temperature inside the food, increasing risk of survival of pathogens like *Clostridium botulinum* during pressure canning of low-acid foods.

Source: PNW 361 Canning meat poultry game

☐ **natural cooling** (process)

Kids: Letting jars get cool down slowly after cooking so they stay safe.

Tweens: Turning off the heat and letting the canner cool down slowly with jars inside before opening it ensures safety.

Teens: Allowing the pressure canner to depressurize and cool naturally before opening prevents unsafe jar breakage and harmful temperature drops.

Adult: The step of allowing a pressure canner to cool and depressurize on its own after processing, included in total processing time, ensuring residual heat continues destroying pathogens in low-acid canned foods.

Source: PNW 361 Canning meat poultry game

☐ **pint jars** (equipment)

Kids: A jar that you can hold easily with one hand to keep food in.

Tweens: A common size glass jar used to hold small amounts of canned food safely.

Teens: A standardized canning jar size, usually 1 pint (16 ounces), used for home canning low-acid foods in appropriate portions.

Adult: One-pint (16 fluid ounce) glass canning jars designed for home food preservation, often recommended for processing low-acid foods in pressure canners because of manageable heating and cooling times.

Source: PNW 421 Use care operation of your pressure canner

☐ **Clostridium botulinum** (safety)

Kids: Tiny germs that can make people very sick if they grow in food.

Tweens: A bacteria that grows in low acid, canned foods and can cause serious sickness.

Teens: A bacterium that produces deadly botulinum toxin, which grows in improperly processed low-acid canned foods.

Adult: *Clostridium botulinum* is an anaerobic, spore-forming bacterium that produces botulinum toxin in low-acid, oxygen-free canned foods if not processed correctly, causing potentially fatal botulism.

Source: PNW 361 Canning meat poultry game

☐ **herbs and vegetables in oil** (safety)

Kids: Plants like garlic or peppers put in oil that need to be kept safe.

Tweens: Low-acid plants stored in oil can let bad germs grow if not kept cold or acidified.

Teens: Storing low-acid herbs or vegetables in oil creates an oxygen-free place that can let *Clostridium botulinum* grow unless acidified or refrigerated properly.

Adult: Low-acid herbs and vegetables stored in oil create an anaerobic environment favorable for *Clostridium botulinum* growth and must be acidified or refrigerated to prevent botulism.

Source: SP 50-701 Herbs and vegetables in oil

☐ **pressure canning** (process)

Kids: A way to cook food in jars using very hot steam to keep it safe.

Tweens: A special cooking step that uses steam under pressure to make low acid foods safe by killing bad germs.

Teens: A method that uses a sealed pot to heat food above boiling temperatures, killing bacteria in low acid foods.

Adult: Pressure canning uses a sealed pressure canner to heat low acid foods to 240°F, destroying *Clostridium botulinum* spores and ensuring safety.

Source: PNW 361 Canning meat poultry game

PICKLING

☐ **vinegar** (ingredient)

Kids: A sour liquid that helps keep pickles tasty and safe.

Tweens: A strong sour juice used to make pickles last longer.

Teens: An acid (usually 5% acetic acid) that prevents bad bacteria in pickling.

Adult: A liquid containing about 5% acetic acid used in pickling to acidify the product and inhibit microbial growth.

Source: PNW 355 Pickling vegetables

☐ **brine** *(ingredient)*

Kids: A salty water that helps make pickles.

Tweens: Salt water that helps pickle and preserve veggies.

Teens: A salt solution used in pickling to extract flavors and preserve vegetables.

Adult: A solution of salt and sometimes vinegar in water used to pickle vegetables by fermenting or preserving.

Source: PNW 355 Pickling vegetables

☐ **fermentation** *(process)*

Kids: A process that makes food taste tangy and last longer.

Tweens: Food changes to sour by good tiny helpers called microbes.

Teens: A natural process where microbes convert sugars into acids to preserve food.

Adult: An anaerobic microbial process converting sugars to organic acids, lowering pH, and preserving vegetables.

Source: PNW 355 Pickling vegetables

☐ **pickling lime** *(ingredient)*

Kids: A special powder that helps make pickles crunchy.

Tweens: A powder added to make pickles firm and crisp.

Teens: Calcium hydroxide used to soak cucumbers to firm their texture before pickling.

Adult: Food-grade calcium hydroxide used to soak cucumbers, increasing firmness before pickling; requires careful rinsing.

Source: PNW 355 Pickling vegetables

☐ **headspace (jar)** *(process)*

Kids: Space left in the jar so the lid can close tight.

Tweens: Space between the food or liquid and the jar lid to seal properly.

Teens: The void space left at the top of a jar to allow for expansion and sealing.

Adult: The gap between packed food or liquid and jar lid, critical for expansion and vacuum during processing.

Source: PNW 355 Pickling vegetables

☐ **pickle spices** *(ingredient)*

Kids: Special powders and seeds that make pickles taste yummy.

Tweens: Spices like dill, mustard, and pepper used to flavor pickles.

Teens: A mix of spices including celery seed, mustard seed, and turmeric added to brine to flavor pickles.

Adult: A blend of spices such as mustard seed, celery seed, turmeric, and sometimes cinnamon, added to pickling brine for flavor.

Source: PNW 355 Pickling vegetables

☐ **crock** *(equipment)*

Kids: A big bowl to hold vegetables when they soak in salty water.

Tweens: A large container used to hold vegetables while they ferment in salty water.

Teens: A 1- to 5-gallon stoneware or food-grade container used for fermenting pickles by holding vegetables submerged in brine.

Adult: A food-grade, often stoneware container typically 1 to 5 gallons in size used to hold and ferment vegetables under brine for pickling.

Source: PNW 355 Pickling vegetables

☐ **weight** *(equipment)*

Kids: Something heavy that pushes the veggies down in the water.

Tweens: A heavy object used to keep vegetables under the salty water when they are pickling.

Teens: An object, such as a plate or jar filled with water, used in fermentation crocks to keep vegetables submerged below brine to prevent spoilage.

Adult: A heavy item like a dinner plate or jars filled with water placed on fermenting vegetables to maintain submersion in the brine, preventing exposure to air and contamination.

Source: PNW 355 Pickling vegetables

☐ **mixed pickling spice** *(ingredient)*

Kids: A special kind of crunchy stuff that makes pickles taste yummy.

Tweens: A mix of seeds and spices added to pickles for flavor.

Teens: A combination of various spices like mustard seed, celery seed, and others mixed together to add traditional flavors in pickling brines.

Adult: A proprietary blend of spices including mustard seed, celery seed, turmeric, and sometimes cinnamon and fennel, added to pickling brines to flavor vegetables during processing.

Source: PNW 355 Pickling vegetables

JAMS & JELLIES

☐ **gel stage** (process)

Kids: It's when jam gets thick and jiggly like jelly.

Tweens: The time when fruit juice and sugar thicken to make jam set.

Teens: The point during cooking when pectin forms a network that traps water, thickening the jam or jelly.

Adult: The gel stage is reached when the jam or jelly mixture thickens and sets due to pectin and sugar interaction, typically tested between 220-221°F or by sheet test.

Source: SP 50-632 Making berry syrups at home

☐ **sugar** (ingredient)

Kids: Sweet white stuff that makes jam taste yummy and helps it get thick.

Tweens: A sweet ingredient that helps fruit jam set and tastes really good.

Teens: A preservative and gelling agent in jams that interacts with pectin and acid to form a gel structure.

Adult: Sucrose added to jams and jellies not only sweetens but promotes gel formation by binding water and interacting with pectin at specified ratios.

Source: SP 50-632 Making berry syrups at home

☐ **acid** (ingredient)

Kids: A sour thing like lemon juice that helps jam get thick and safe.

Tweens: Sour juice that helps jams and jellies set and stops germs from growing.

Teens: Acid (like lemon juice) lowers pH to make gel set properly and prevents bacterial growth in jams and jellies.

Adult: Acidification by adding lemon juice or citric acid lowers jam pH below 4.6, necessary for pectin gelation and safe preservation by inhibiting pathogens.

Source: SP 50-632 Making berry syrups at home

☐ **foam** (process)

Kids: Bubbles that sometimes float on the top of hot jam like bubbles in a bath.

Tweens: Soft bubbles that form on jam while cooking; usually taken off before canning.

Teens: Impurities and air create foam on boiling jam surface, which is skimmed off to improve clarity and preservation.

Adult: Foam is the frothy layer containing air and solids forming during jam cooking; it should be skimmed off before jar filling to prevent spoilage.

Source: SP 50-632 Making berry syrups at home

☐ **sterilized jars** (process)

Kids: Clean jars made very hot to keep bugs and germs away.

Tweens: Jars that are boiled or heated so no germs can grow inside.

Teens: Glass jars treated by boiling or hot water canning to kill microorganisms before filling.

Adult: Pre-sterilized jars, boiled for at least 10 minutes in water, eliminate contaminants ensuring a safe seal and shelf stability for jams.

Source: SP 50-632 Making berry syrups at home

☐ **uncooked freezer jam** (process)

Kids: Jam made without cooking, like a cold, sweet fruit jelly.

Tweens: Jam made by mixing fruit and pectin without boiling, then freezing it.

Teens: Jam made by blending crushed fruit, sugar and hot pectin, then cooled and frozen to avoid cooking.

Adult: Uncooked freezer jams use powdered pectin mixed with sugar and crushed fruit at room temperature, then stored frozen, retaining fresh fruit flavor and color.

Source: SP 50-632 Making berry syrups at home

☐ **gel point** (result)

Kids: This is when jam or jelly becomes thick and sticky like a soft candy.

Tweens: The gel point is when the mixture starts to get thick and hold its shape.

Teens: The gel point is the temperature or stage when the jam or jelly thickens enough to form a gel network.

Adult: Gel point is the stage in jam or jelly preparation when the mixture reaches a temperature about 8°F above boiling water and thickens sufficiently to form a gel, indicated by the mixture 'sheeting' from a spoon.

Source: SP 50-632 Making berry syrups at home

☐ **jelly bag** (equipment)

Kids: A bag that helps catch only the juice from smashed fruit.

Tweens: A special cloth bag used to strain jelly juice from fruit pulp.

Teens: A reusable porous bag designed for extracting clear juice by straining cooked fruit, leaving seeds and pulp behind.

Adult: A permeable fabric bag used to drain crushed fruit mixture for jelly extraction, allowing liquid juice to drip through while retaining solids, essential for clear jelly preparation.

Source: SP 50-632 Making berry syrups at home

☐ **low-methoxyl pectin** (ingredient)

Kids: A special stuff from fruits that helps jams get thick without lots of sugar.

Tweens: A type of natural ingredient that makes jam thick even with less or no sugar.

Teens: A pectin variant that sets with calcium ions and requires less or no sugar, useful for low-sugar preserves.

Adult: A type of pectin with fewer methoxyl groups that forms gels in the presence of calcium ions without requiring high sugar or acid levels, enabling low- or no-sugar fruit spreads safely processed and stored.

Source: SP 50-632 Making berry syrups at home

FREEZING & COLD STORAGE

☐ **freezer burn** (safety)

Kids: When food in the freezer gets dry and yucky on the edges.

Tweens: Freezer burn happens when frozen food loses water and gets tough and changes color or smell.

Teens: Freezer burn is surface dehydration on frozen food caused by moisture loss, leading to tough texture and off flavor.

Adult: Freezer burn is dehydration and oxidation damage from moisture loss when ice crystals sublime during freezer temperature fluctuations, causing texture and flavor changes.

Source: PNW 586 Home freezing seafood

☐ **vacuum packaging** (process)

Kids: A way to keep food safe by removing air from a bag.

Tweens: Vacuum packaging means sealing food in bags after taking all the air out to keep it fresh longer.

Teens: Vacuum packaging removes air to reduce oxidation and slows freezer burn by limiting ice crystal growth and moisture loss.

Adult: Vacuum packaging involves extracting air from packaging to limit oxidation, reduce freezer burn, and preserve frozen food quality by preventing moisture migration and large ice crystals.

Source: PNW 586 Home freezing seafood

☐ **freezer glaze** (process)

Kids: A thin ice coat on frozen fish to keep it fresh.

Tweens: Freezer glaze is water frozen onto seafood to protect it from drying out in the freezer.

Teens: Freezer glaze is a protective ice coating formed by dipping frozen seafood in cold water to prevent freezer burn and oxidation.

Adult: Freezer glaze is a thin coating of ice applied to frozen seafood by dipping in cold water, serving as a barrier to oxygen and moisture loss during storage.

Source: PNW 586 Home freezing seafood

☐ **cling plastic wrap** (equipment)

Kids: A clear, sticky plastic that wraps food tight.

Tweens: Cling plastic wrap is stretchy plastic used to cover food to keep it fresh and stop air from getting in.

Teens: Cling plastic wrap is a thin, stretchable plastic film used as a final wrap for freezing to reduce air exposure and protect foods.

Adult: Cling plastic wrap is a flexible film material that adheres tightly to food surfaces to exclude air and reduce oxidation and dehydration in freezing applications.

Source: PNW 586 Home freezing seafood

☐ **glazing** (process)

Kids: Covering frozen fish with a thin ice coat.

Tweens: Glazing means putting a thin layer of ice on frozen food to protect it from freezer burn.

Teens: Glazing is dipping frozen seafood in cold water to coat it with ice, creating a protective barrier during storage.

Adult: Glazing involves dipping frozen food in cold water to form a protective ice layer that helps prevent dehydration and oxidation in frozen storage.

Source: PNW 586 Home freezing seafood

☐ **freezer temperature** (safety)

Kids: How cold the freezer is to keep food frozen.

Tweens: Freezer temperature is how cold the freezer must be to keep food safe and frozen.

Teens: Freezer temperature should be 0°F (-18°C) or lower to safely preserve food and minimize quality loss.

Adult: Freezer temperature must be maintained at 0°F (-18°C) or below to safely inhibit microbial growth and maintain frozen food quality.

Source: PNW 586 Home freezing seafood

☐ **master bag** *(equipment)*

Kids: A bigger bag that holds smaller food packages.

Tweens: A master bag is a big freezer bag used to put lots of smaller wrapped foods inside.

Teens: A master bag is a heavy-duty freezer bag used to group smaller individually wrapped frozen items for easier storage and freezing.

Adult: A master bag is a large polyethylene freezer bag intended to hold multiple individually wrapped units, minimizing air exposure and simplifying handling.

Source: PNW 586 Home freezing seafood

☐ **air exclusion** *(process)*

Kids: Keeping air away from food so it stays fresh longer.

Tweens: Air exclusion means keeping as much air out of the package as you can to stop food from drying out or going bad.

Teens: Air exclusion is removing or limiting air inside packaging to reduce oxidation and ice crystal growth that damage frozen food.

Adult: Air exclusion involves packaging methods that minimize air presence to limit oxidation, reduce ice crystal formation, and prevent freezer burn in frozen foods.

Source: PNW 586 Home freezing seafood

☐ **labeling** *(process)*

Kids: Putting a sticker or mark on food to say what's inside and when it was frozen.

Tweens: Labeling means writing what the food is and when you froze it so you remember later.

Teens: Labeling frozen food packages with date, contents, and weight helps keep track and avoid waste from forgotten packages.

Adult: Labeling packages with date, product identity, weight, and storage location maintains inventory control and prevents waste in frozen food management.

Source: PNW 586 Home freezing seafood

DEHYDRATING & SMOKING

☐ **pellicle** *(process)*

Kids: A sticky shiny skin that forms on meat before smoking.

Tweens: A sticky coat on fish that helps smoke stick and keeps it safe.

Teens: A tacky surface layer on fish formed during drying that helps smoke absorption and prevents spoilage.

Adult: A slightly tacky skin formed on meat or fish surface during drying; it allows even smoke deposition and inhibits surface spoilage during smoking.

Source: PNW 238 Smoking fish at home

☐ **brining** *(process)*

Kids: Soaking food in salty water before cooking or smoking.

Tweens: Putting food in salty water to add flavor and help keep it safe.

Teens: Soaking fish or meat in a saltwater solution to add flavor and guard against bacteria before smoking.

Adult: Soaking fish or meat in a saltwater solution (e.g., 1 cup salt to 7 cups water) to add moisture, flavor, and inhibit spoilage before smoking.

Source: PNW 238 Smoking fish at home

☐ **smokehouse** *(equipment)*

Kids: A small house where food gets smoky to help keep it fresh.

Tweens: A special place to hang food and add smoke to preserve it.

Teens: An enclosed structure that heats wood or chips to produce smoke for curing and flavoring meats or fish.

Adult: An enclosed unit or building designed for controlled exposure of foods to smoke from burning wood or chips for preservation and flavor enhancement.

Source: SP 50-921 Drying herbs

☐ **dehydrator** *(equipment)*

Kids: A machine that takes water out of food to make it dry.

Tweens: A machine that dries food by blowing warm air to preserve it.

Teens: An appliance that removes moisture from foods using warm air circulation to preserve them.

Adult: An electric or gas-powered appliance that removes moisture from food through controlled warm air circulation to make shelf-stable dried food products.

Source: SP 50-921 Drying herbs

☐ **hot smoking** *(process)*

Kids: Cooking food with warm smoke to make it tasty and safe.

Tweens: Using warm smoke that cooks the food to keep it safe and yummy.

Teens: Smoking food with heat high enough to cook it and kill bacteria, usually above 150°F internal temperature.

Adult: A smoking process where fish or meat is exposed to smoke and heat to cook it and reach an internal temperature of at least 150°F for safety.

Source: PNW 238 Smoking fish at home

☐ **cold smoking** *(process)*

Kids: Adding smoke to food without cooking it, to help keep it safe.

Tweens: Smoking food with cool smoke to add flavor but not cooking it.

Teens: Exposing food to smoke at low temperatures (under 90°F) without cooking, often used before cooking or drying.

Adult: A smoking method where food is exposed to smoke at temperatures below 90°F to add flavor without cooking; often requires further cooking for safety.

Source: PNW 238 Smoking fish at home

☐ **salometer** *(equipment)*

Kids: A tool that checks how salty water is for soaking food.

Tweens: A device that measures salt levels in brine before soaking food.

Teens: An instrument used to measure salt concentration in brining solutions for smoking and curing.

Adult: A scientific instrument that measures the salinity of brine solutions, used to ensure proper salt concentration for curing and smoking.

Source: PNW 238 Smoking fish at home

☐ **jerky** *(result)*

Kids: Dry chewy meat you can eat like a snack.

Tweens: Meat that is dried and thin so you can chew it as a treat.

Teens: Lean meat dried at low temperatures after seasoning, preserved for long storage without refrigeration.

Adult: Lean strips of meat cured and dried to reduce moisture content to below 20%, inhibiting bacteria growth and allowing shelf stability.

Source: PNW 632 Making jerky at home safely

☐ **drying** *(process)*

Kids: Taking water out of food to keep it safe for a long time.

Tweens: Removing moisture from food so it won't spoil quickly.

Teens: The process of removing moisture from foods to prevent bacteria growth and spoilage.

Adult: The preservation technique that removes sufficient moisture from foods by heat, air or sun drying to inhibit microbial growth and enzymatic spoilage.

Source: PNW 397 Drying fruits and vegetables